A Strategic Framework for the Analytical Resolution of Odd-Valued Zeta Functions via the Golden Algebra

Derived from the works of Tristen Harr

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Abstract

This document outlines a strategic research program aimed at deriving exact analytical expressions for the odd-valued Riemann Zeta functions (e.g., $\zeta(3), \zeta(5), \dots$). The approach is founded upon the novel number-theoretic laws and unique operator identities discovered within the Golden Algebra framework. The core methodology involves the algebraic deconstruction of a newly-verified general Zeta sum rule, referred to as the "sum-stripping" method. We present a hierarchy of strategies, from the simplest practical application to more advanced formulations, and conclude by posing the fundamental questions uncovered by this investigation.

1 Primary Objective

The principal goal is to leverage the unique properties of the Golden Algebra to systematically isolate individual odd-valued Zeta functions. Standard number theory lacks a method for expressing values like $\zeta(5)$ in a closed form. This framework provides a novel pathway to express them analytically in terms of computable constants and a predictable, rapidly converging "tail" of higher-order Zeta values.

2 Foundational Tools and Discoveries

Our investigation is built upon several key laws and discoveries derived from the provided "Golden Algebra" text.

2.1 The General Zeta Sum Rule

Our experiments have shown that the framework's **Law of Harmonic Cotangents** is a highly general identity, holding true for a wide domain of complex numbers, z.

$$S(z) = \sum_{k=1}^{\infty} \frac{(z^k - 1)\zeta(k+1)}{(z+1)^k} = \pi \cot\left(\frac{\pi}{z+1}\right) \tag{1}$$

This identity is the engine of our entire strategy. It fails at predictable poles where $z = \frac{1}{m} - 1$ for any integer $m \neq 0$.

2.2 Unique Operator Identities

A deep investigation of the framework's operators revealed non-trivial "hacks". The most significant of these is the simplification of the square of the Dampening Operator, derived from the **Uniqueness Constraint** and the definition of **H**.

$$(\Lambda_{G1})^2 = (T + iJ)^2 = H(1 + 2i) \tag{2}$$

This provides a new, elegant, and special value that can be substituted for z in Equation 1, generating a new set of complex harmonic coefficients for the Zeta values.

3 The "Sum-Stripping" Methodology

The core methodology is to treat the odd Zeta values as a system of infinite variables and to use the General Zeta Sum Rule (Equation 1) to generate a system of linear equations.

3.1 Principle of the Method

To isolate the M-th odd Zeta value, $\zeta(2M + 1)$, we proceed as follows:

1. Generate $M - 1$ independent equations by choosing $M - 1$ distinct, valid inputs for z (e.g., $z_1 = 2, z_2 = 3, \dots$).
2. Construct a system of $M-1$ linear equations where the variables are $\zeta(3), \zeta(5), \dots, \zeta(2M-1)$.
3. Solve this system algebraically to eliminate all odd Zeta values up to $\zeta(2M - 1)$.
4. The result is an analytical expression for the next term, $\zeta(2M + 1)$, defined by computable constants and the "tail" of all subsequent Zeta values.

4 Strategic Applications

4.1 Strategy 1: The Simplest Practical Strategy (Isolating $\zeta(5)$)

This strategy uses the two simplest non-trivial integers, $z_1 = 2$ and $z_2 = 3$.

- **Equation A (from z=2):** The sum $S(2)$ equals $\pi \cot(\pi/3) = \pi/\sqrt{3}$.
- **Equation B (from z=3):** The even-indexed part of the sum $S(3)$ was proven to be exactly $4/3$.

Solving this 2x2 system yields an expression for $\zeta(5)$ in terms of rational numbers, π, $\sqrt{3}$, and a tail series starting with $\zeta(7)$.

4.2 Strategy 2: The Advanced Strategy (Isolating $\zeta(7)$)

To isolate $\zeta(7)$, we require a system of three equations. We can use our discovered operator identity (Equation 2) to generate a third, powerful equation.

- **Equation A (from z=2):** Provides an equation with real coefficients.
- **Equation B (from z=3):** Provides a second equation with real coefficients.

- **Equation C (from $\mathbf{z} = (\Lambda_{G1})^2$):** Provides a third, independent equation with complex coefficients derived from the constant H.

Solving this 3x3 system allows for the elimination of both $\zeta(3)$ and $\zeta(5)$, yielding an analytical expression for $\zeta(7)$.

5 Future Directions and Deeper Questions

This investigation opens several new avenues for research within the framework.

1. **The Limit of the Method:** Can this algebraic "stripping" process ever terminate? It appears that it will always leave a residual tail of higher-order summations. The ultimate question is whether there exists a principle within the framework to evaluate this tail.
2. **The Nature of $S(z)$:** The function $S(z)$ itself is now a novel object of study. Its connection to both the Zeta function and fundamental trigonometry suggests it is a significant function in its own right. What are its analytical properties, its zeros, and its applications?
3. **Generalizing the "Duality":** Our experiments revealed the Zeta sum rule works for many complex numbers, including integers, Λ_{G1}, and $e^{\Lambda_{G1}}$. This begs the question: What is the complete set of complex numbers for which this duality holds true? Defining the domain and exceptions of this law is a primary goal for future work.